ME551 Vibrations in Linear Systems

Lecture 7

Single Degree of Freedom Systems 1

Chap 3. Single-Degree-of-Freedom System

3.1 Response to Initial Excitations
3.2 Response to Harmonic Excitations
3.3 Systems with Structural Damping
3.4 Response to Periodic Excitations
3.5 Response to Arbitrary Excitations
3.6 Discrete-Time Systems (optional)

$\boldsymbol{F}_i + f_i + \sum_{j=1, j\neq i}^{N} \boldsymbol{f}_{ij} = m_i \ddot{\boldsymbol{r}}_i$: Eq. of motion for a particle in a system of particles

$$\downarrow \Leftarrow \begin{cases} x \Leftarrow \boldsymbol{r}_i : \text{ linear motion for } N = 1 \\ -c\dot{x} - kx + f(t) \Leftarrow \boldsymbol{F}_i + f_i + \sum_{j=1, j\neq i}^{N} \boldsymbol{f}_{ij} \end{cases}$$

$$m\ddot{x} + c\dot{x} + kx = f(t) \begin{cases} 3.1: x(0) \,\&\, \dot{x}(0), f(t) = 0 \\ 3.2: f(t) = f_0 e^{i\omega t} \\ 3.3: c = \dfrac{\alpha}{\pi\omega}, f(t) = f_0 e^{i\omega t} \\ 3.4: f(t) = f(t+T) \end{cases}$$

3.1 Response to Initial Condition (or Excitation) 1st Order System

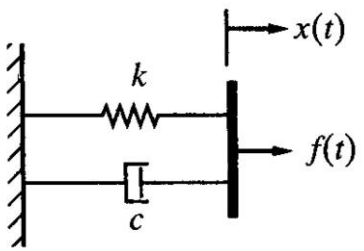


Figure 3.1 Damper-spring system

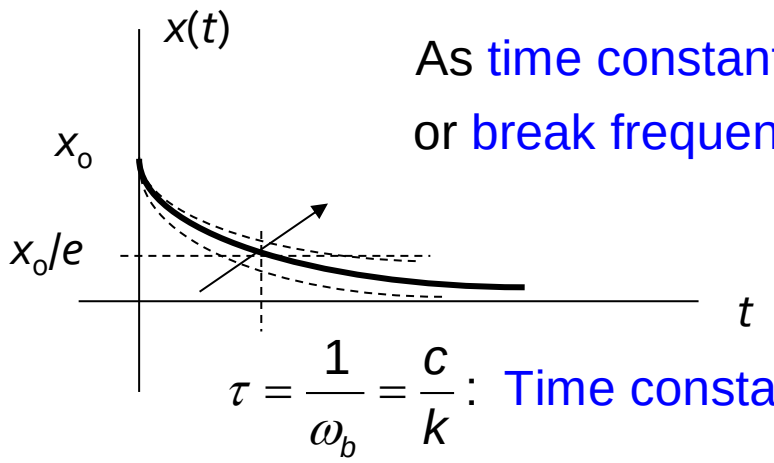
$c\dot{x} + kx = 0$ with $x(0^+) = x_0 \left[\Leftrightarrow c\dot{x} + kx = cx_0\delta(t), x(0^-) = 0,\ \delta(t): \text{Dirac delta function} \right]$

$\downarrow \Leftarrow$ Laplace transform: $L\left[\dot{x}(t)\right] = sX(s) - x(0) \quad \Leftarrow L\left[\dfrac{d^r x(t)}{dt^r}\right] = s^r X(s) - s^{r-1}x(0) - s^{r-2}\dot{x}(0) - s^{r-3}\ddot{x}(0) - \ldots$

$$X(s) = \frac{cx_0}{cs + k} = \frac{x_0}{s + \omega_b}, \text{where } \omega_b = \frac{k}{c} \equiv \frac{1}{\tau}\left(\frac{\text{N/m}}{\text{N/(m/sec)}} = \text{radian/sec} \right)$$

$\downarrow \Leftarrow$ Inverse Laplace transform: $L^{-1}\left[\dfrac{1}{s-p}\right] = e^{pt}$

$$x(t) = x_0 e^{-\omega_b t} \text{ for } t \geq 0$$



As time constant τ increases

or break frequency ω_b decreases

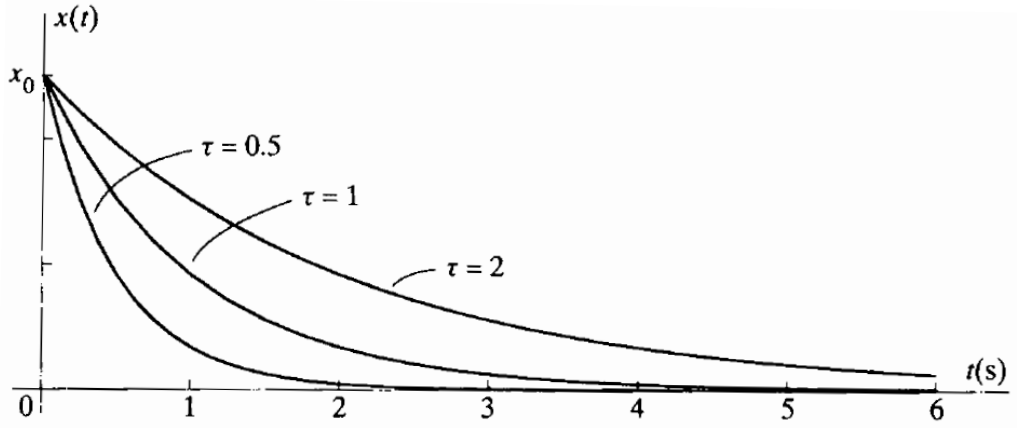


Figure 3.2 Free response of a damper-spring system with the time constant as a parameter

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Response to Initial Condition (or Excitation) 2nd Order System

$$m\ddot{x}+c\dot{x}+kx=0 \text{ with } \begin{cases} x(0)=x_0 \\ \dot{x}(0)=\dot{x}_0 \end{cases} \qquad \left[\Leftrightarrow m\ddot{x}+c\dot{x}+kx=m\dot{x}_0\delta(t) \text{ with } \begin{cases} x(0)=x_0 \\ \dot{x}(0^-)=0 \end{cases} \right]$$

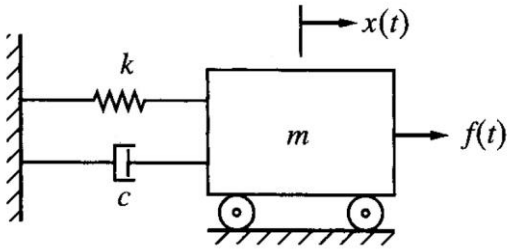

Figure 3.3 Damped single-degree-of-freedom system

$\downarrow\Leftarrow$ Laplace transform: $L\left[\ddot{x}(t)\right]=s^2X(s)-sx(0)-\dot{x}(0),\ L\left[x(t)\right]=sX(s)-x(0)$

$$X(s)=\frac{mx_0s+m\dot{x}_0+cx_0}{ms^2+cs+k}=\frac{x_0s+\dot{x}_0+\frac{c}{m}x_0}{s^2+\frac{c}{m}s+\frac{k}{m}}=\frac{B}{s-p}+\frac{B^*}{s-p^*}$$

$\downarrow\Leftarrow$ Inverse Laplace transform: $L^{-1}\left[\frac{1}{s-p}\right]=e^{pt}$

$$x(t)=Be^{pt}+B^*e^{p^*t}=Ae^{-\varsigma\omega_n t}\cos(\omega_d t-\phi), \text{ where}$$

$\omega_n=\sqrt{\frac{k}{m}}$: undamped (circular) natural frequency [rad/sec], $f_n=\frac{\omega_n}{2\pi}$ [Hz]

$\varsigma=\frac{c}{2\sqrt{mk}}$: damping ratio, $\omega_d=\omega_n\sqrt{1-\varsigma^2}$: damped (circular) natural frequency

$$p=-\varsigma\omega_n+j\omega_d, B=\frac{x_0}{2}-j\frac{\dot{x}_0+\frac{c}{2m}x_0}{2\omega_d}$$

$$A=\frac{1}{\omega_d}\sqrt{(\omega_d x_0)^2+(\varsigma\omega_n x_0+\dot{x}_0)^2}, \phi=\tan^{-1}\left(\frac{\varsigma\omega_n x_0+\dot{x}_0}{\omega_d x_0}\right)$$

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Response to Initial Excitation(or Condition) 2nd Order System

Undamped system

$$m\ddot{x} + kx = 0$$

$$\ddot{x} + \omega_n^2 x = 0$$

$$x(t) = A\cos(\omega_n t - \phi)$$

Non-dimensional parameters

$$\underbrace{\omega_n = \sqrt{\frac{k}{m}}}_{\text{Undamped Natural Frequency}}$$

$$\underbrace{\varsigma = \frac{c}{2\sqrt{mk}}}_{\text{Damping Ratio}}$$

$$\underbrace{\omega_d = \omega_n\sqrt{1-\varsigma^2}}_{\text{Damped Natural Frequency}}$$

Damped system

$$m\ddot{x} + c\dot{x} + kx = 0$$

$$\ddot{x} + 2\varsigma\omega_n\dot{x} + \omega_n^2 x = 0$$

$$x(t) = \underbrace{Ae^{-\varsigma\omega_n t}}_{\text{Envelope}}\underbrace{\cos(\omega_d t - \phi)}_{\text{Periodic Term}}$$

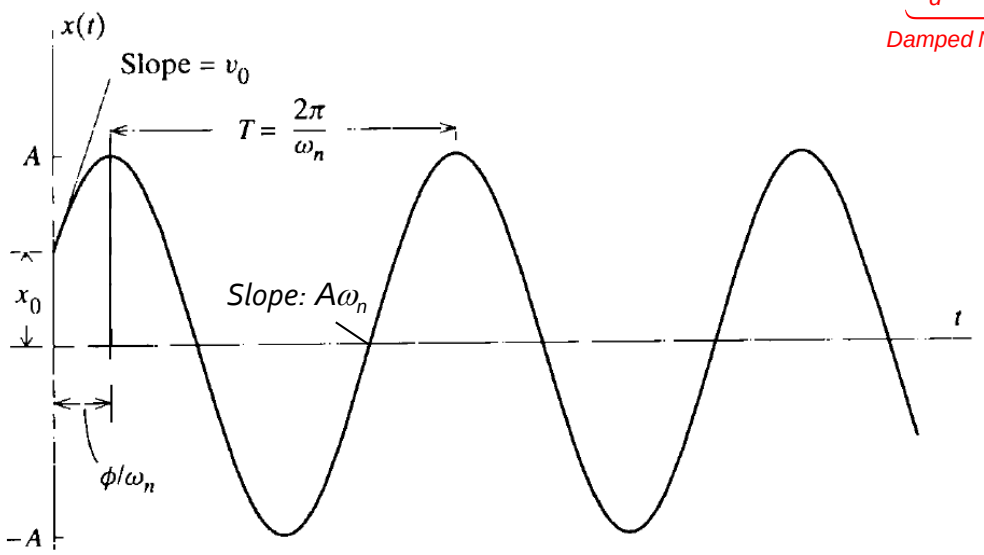

Figure 3.4 Free response of an undamped single-degree-of-freedom system

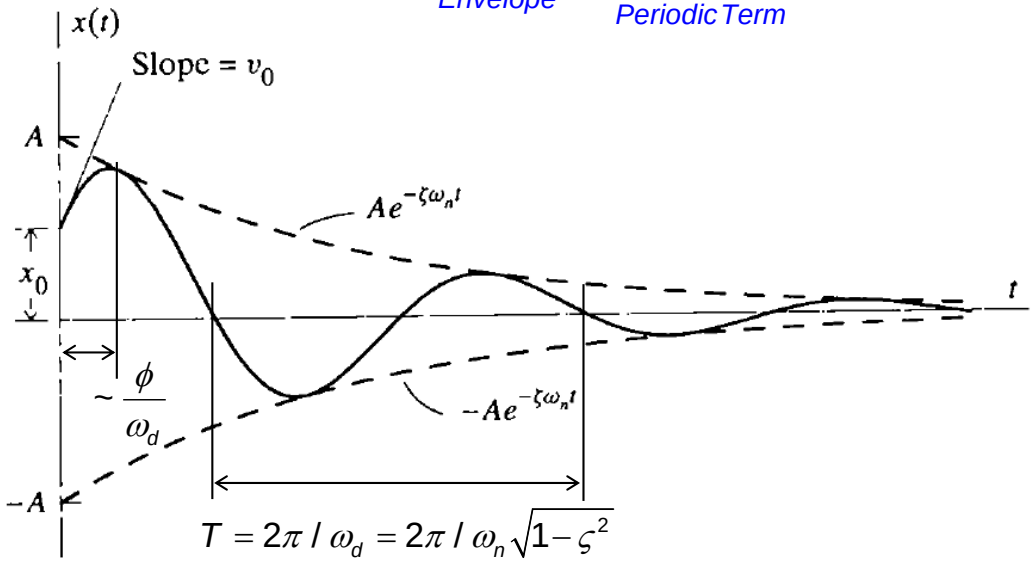

Figure 3.5 Free response of an underdamped single-degree-of-freedom system

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3.2 Response to Harmonic Excitations 1st Order System

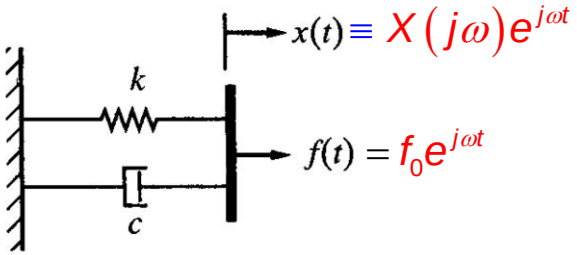

Figure 3.1 Damper-spring system

$$c\dot{x} + kx = f(t) = f_0 e^{j\omega t}$$

$$\downarrow \Leftarrow x(t) = x_0 e^{j(\omega t - \phi)} \equiv X(j\omega)e^{j\omega t}$$

$$G(j\omega) \equiv \frac{X(j\omega)}{f_0} = \frac{x_0 e^{-j\phi}}{f_0} = \frac{1}{jc\omega + k} = \frac{1}{\sqrt{k^2 + (c\omega)^2}} e^{-j\tan^{-1}\left(\frac{c\omega}{k}\right)}$$: Compliance FRF

$$\downarrow$$

Magnitude & Phase :

$$\begin{cases} |G(j\omega)| = \dfrac{1}{\sqrt{k^2 + (c\omega)^2}} \left(\dfrac{\mathrm{m}}{\mathrm{N}}\right) \Rightarrow |G(j\omega)|_{\omega=0} = \dfrac{1}{k} \\ \phi = \tan^{-1}\left(\dfrac{c\omega}{k}\right) < \dfrac{\pi}{2} \end{cases}$$

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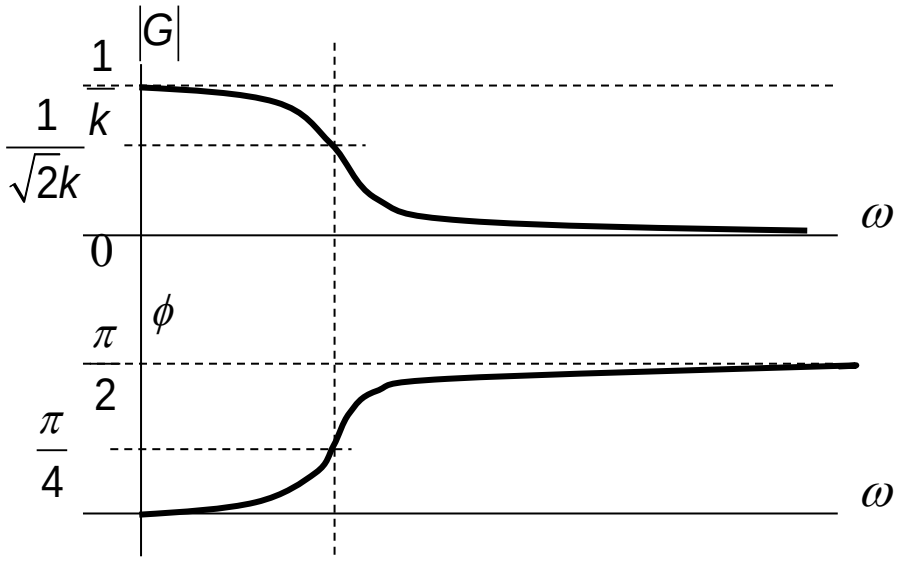
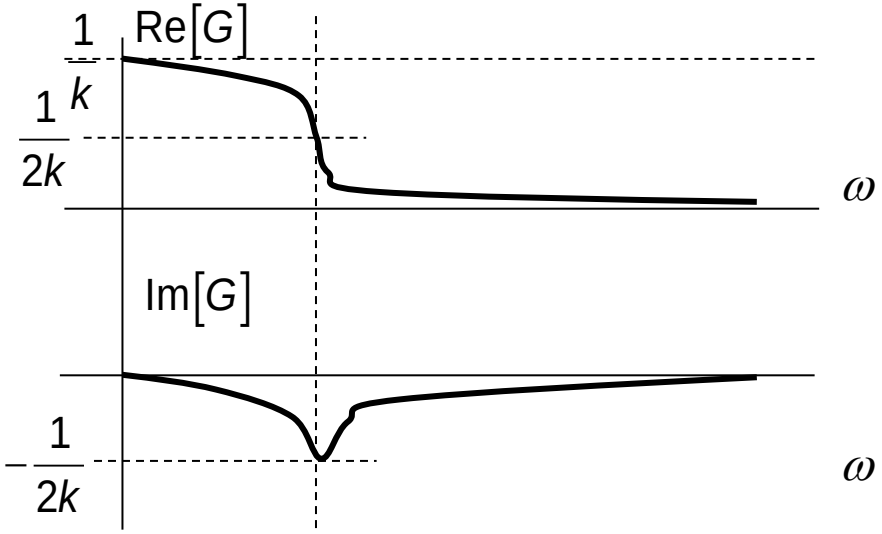
Plots of Compliance FRF 1st Order System

$$G(j\omega) = \frac{1}{\sqrt{k^2 + (c\omega)^2}} e^{-j\tan^{-1}\left(\frac{c\omega}{k}\right)} = \frac{k}{k^2 + (c\omega)^2} - j\frac{c\omega}{k^2 + (c\omega)^2}$$



$$\omega_b = \frac{1}{\tau} = \frac{k}{c}$$



$$\omega_b = \frac{1}{\tau} = \frac{k}{c}$$

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Response to Harmonic Excitations 2nd Order System

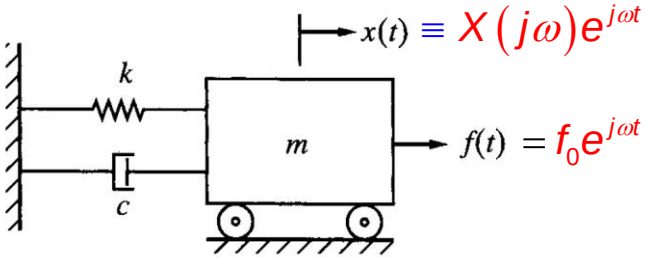

Figure 3.3 Damped single-degree-of-freedom system

$$m\ddot{x} + c\dot{x} + kx = f_0 e^{j\omega t}$$

$$\downarrow \Leftarrow x(t) = X(j\omega)e^{j\omega t}$$

$$G(j\omega) \equiv \frac{X(j\omega)}{f_0} = \frac{1}{-m\omega^2 + jc\omega + k} = \frac{1}{\sqrt{(k-m\omega^2)^2 + (c\omega)^2}} e^{-j\tan^{-1}\left(\frac{c\omega}{k-m\omega^2}\right)}$$: compliance FRF

$$\downarrow$$

Mag & Phase :
$$\begin{cases} |G(j\omega)| = \dfrac{|X(j\omega)|}{f_0} = \dfrac{1}{\sqrt{(k-m\omega^2)^2 + (c\omega)^2}} = \dfrac{1/k}{\sqrt{(1-r^2)^2 + (2\varsigma r)^2}}, \text{ where } r = \dfrac{\omega}{\omega_n} \\ \phi = \tan^{-1}\left(\dfrac{c\omega}{k-m\omega^2}\right) = \tan^{-1}\left(\dfrac{2\varsigma r}{1-r^2}\right) \left[\phi = \dfrac{\pi}{2} \text{ at } \omega = \omega_n\right] \end{cases}$$
: Invariant point

$\mathrm{Re}[G(j\omega)]$ & $\mathrm{Imag}[G(j\omega)]$:
$$\begin{cases} \mathrm{Re}[G(j\omega)] = \dfrac{k-m\omega^2}{(k-m\omega^2)^2 + (c\omega)^2} = \dfrac{1}{k}\dfrac{1-r^2}{(1-r^2)^2 + (2\varsigma r)^2} \\ \mathrm{Im}[G(j\omega)] = -\dfrac{1}{k}\dfrac{2\varsigma r}{(1-r^2)^2 + (2\varsigma r)^2} \end{cases}$$

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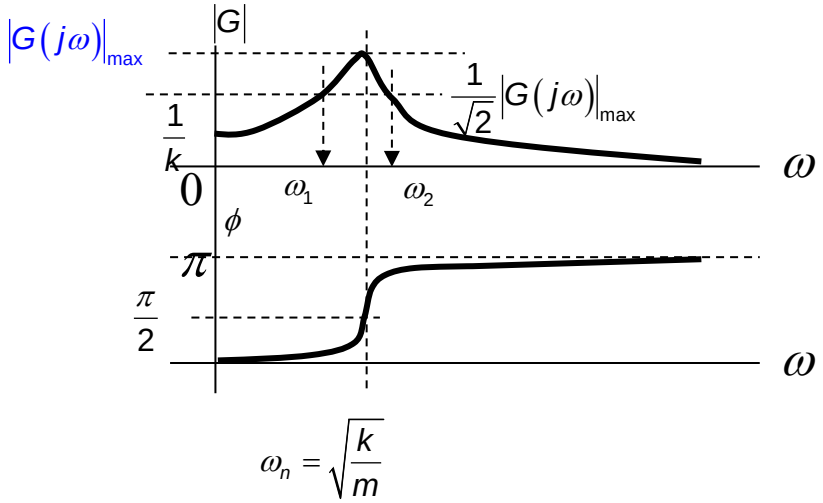
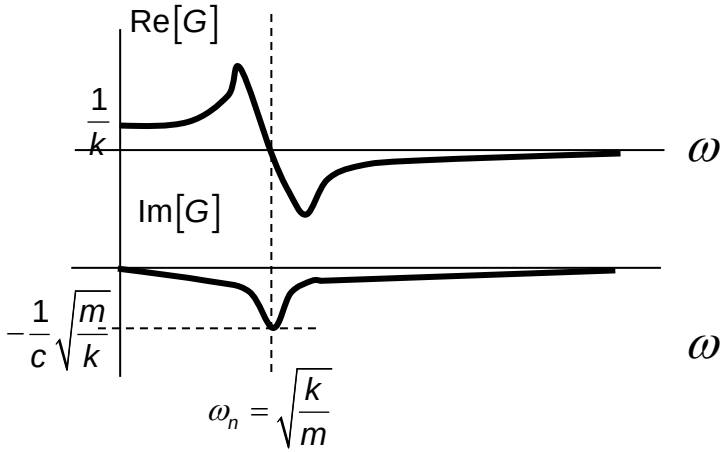
Plots of Compliance FRF 2nd Order System

$$r = \frac{\omega}{\omega_n} : \begin{cases} \left|G(j\omega)\right| = \dfrac{1/k}{\sqrt{\left(1-r^2\right)^2 + \left(2\varsigma r\right)^2}}, \ \phi = \tan^{-1}\left(\dfrac{2\varsigma r}{1-r^2}\right) \\ \mathrm{Re}\left[G(j\omega)\right] = \dfrac{1}{k}\dfrac{1-r^2}{\left(1-r^2\right)^2 + \left(2\varsigma r\right)^2}, \ \mathrm{Im}\left[G(j\omega)\right] = -\dfrac{1}{k}\dfrac{2\varsigma r}{\left(1-r^2\right)^2 + \left(2\varsigma r\right)^2} \end{cases}$$





$\varsigma \cong \dfrac{\omega_2 - \omega_1}{2\omega_n}$, where $\left|G(j\omega)\right|_{\omega_1} = \left|G(j\omega)\right|_{\omega_2} = \dfrac{1}{\sqrt{2}}\left|G(j\omega)\right|_{max}$, ω_1 & ω_2 : Half power frequency points

$Q \equiv \dfrac{\left|G(j\omega)\right|_{max}}{\left|G(j\omega)\right|_{\omega=0}} \cong \dfrac{1/(2k\varsigma)}{1/k} = \dfrac{1}{2\varsigma}$: Quality factor (or dynamic amplification factor)

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Plots of Compliance FRF 2nd Order System w.r.t damping ratio

$$\left|G(j\omega)\right|\cdot k = \frac{1}{\sqrt{\left(1-r^2\right)^2+\left(2\varsigma r\right)^2}}, \text{ where } r = \frac{\omega}{\omega_n} \qquad \phi = \tan^{-1}\left(\frac{2\varsigma r}{1-r^2}\right)$$

: Amplification Factor

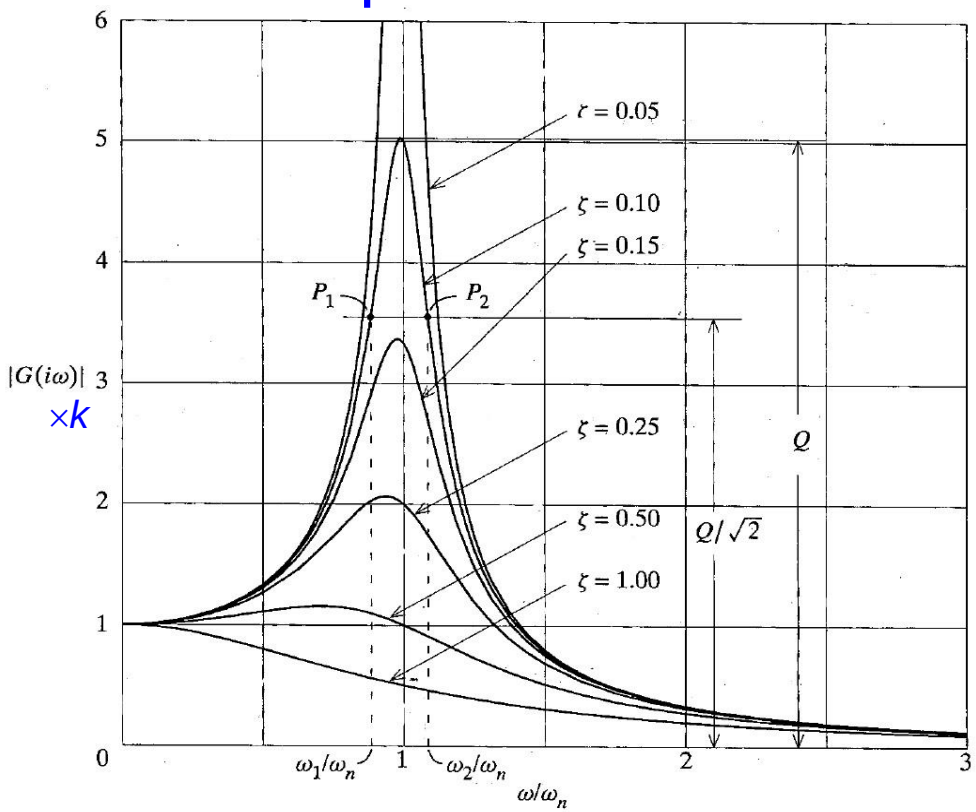

Figure 3.9 Magnitude of the frequency response for a damped single-degree-of-freedom system versus normalized excitation frequency with the damping factor as a parameter

: Phase

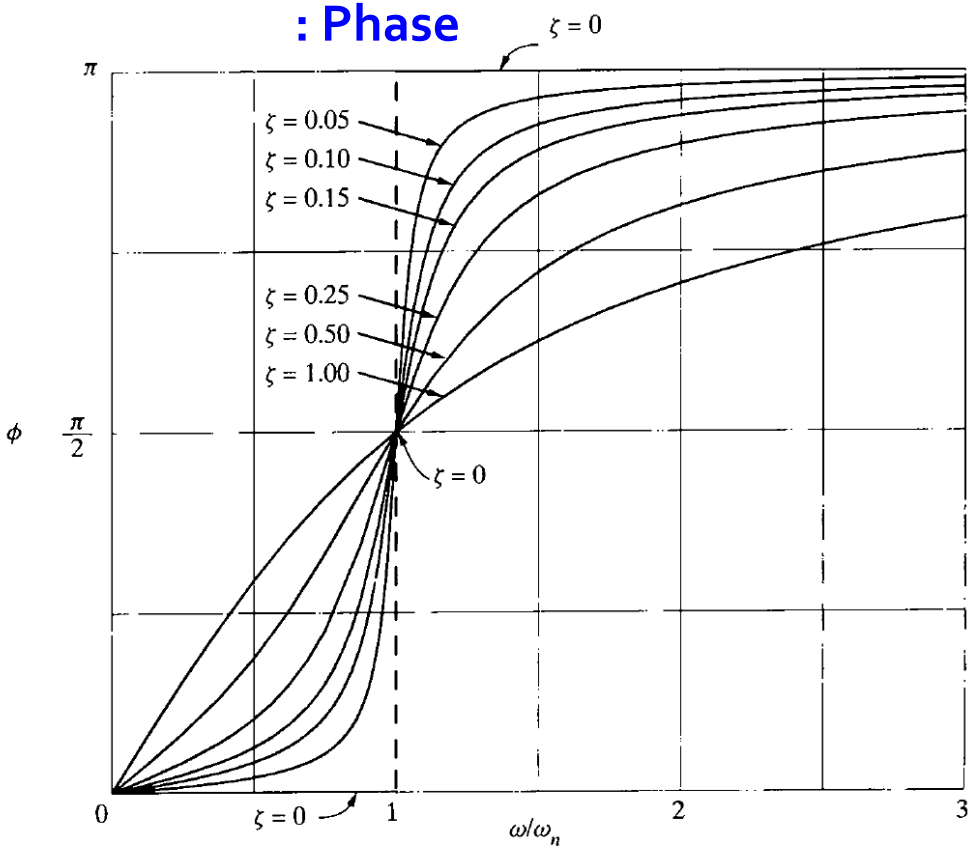

Figure 3.10 Phase angle of the frequency response for a damped single-degree-of-freedom system versus normalized excitation frequency with the damping factor as a parameter

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3.3 Systems with Structural Damping

- Structural damping is generally attributed to energy loss due to the hysteresis of elastic materials experiencing cyclic stress. Mathematically, structural damping is commonly treated as an *equivalent viscous damping*.
- Energy dissipation in a single-DOF system with viscous damping (dissipation/cycle versus dissipation/time)

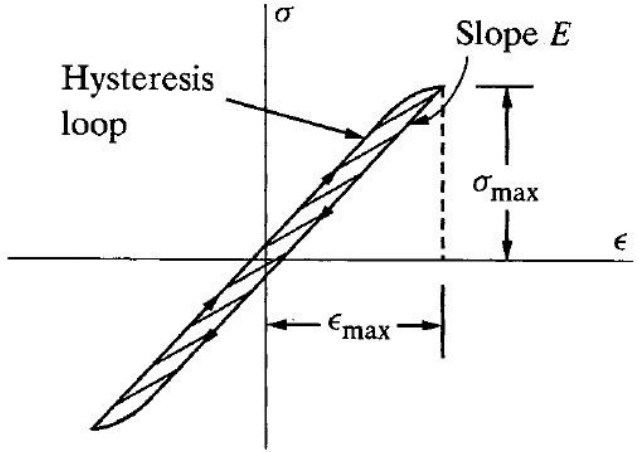

Figure 3.14 Stress-strain diagram showing a hysteresis loop

$$\Delta E_{cycle} \equiv \oint F_d dx = \int_0^{2\pi/\omega} F_d \dot{x} dt$$

$$\downarrow \Leftarrow \begin{cases} m\ddot{x} + c\dot{x} + kx = f_0 e^{j\omega t} \\ \downarrow \\ x(t) = x_0 e^{j(\omega t - \phi)} \end{cases}$$

$$= \oint \mathrm{Re}[c\dot{x}]\mathrm{Re}[dx] = \int_0^{2\pi/\omega} c\,\mathrm{Re}[\dot{x}]\mathrm{Re}[\dot{x}]dt = c(\omega x_0)^2 \int_0^{2\pi/\omega} \cos^2(\omega t - \phi)dt$$

$$= \pi c \omega x_0^2 \Rightarrow c_{eq} = \frac{\Delta E_{cycle}}{\pi \omega x_0^2}$$

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- Energy dissipation per cycle in many materials subject to harmonic loading:

$$\Delta E_{cycle} \propto x_0^2 \text{ or } \Delta E_{cycle} = \alpha x_0^2,\ \alpha\text{: constant} \Rightarrow \text{Structural damping}$$

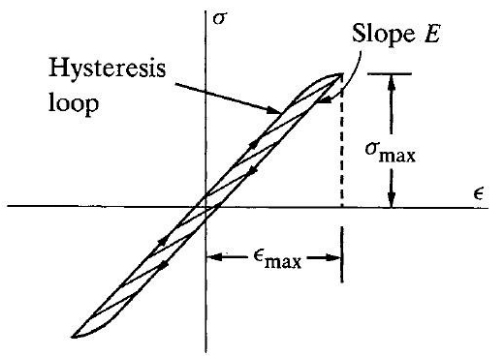

Figure 3.14 Stress-strain diagram showing a hysteresis loop

$$m\ddot{x} + c\dot{x} + kx = f_0 e^{j\omega t}$$

$$\downarrow\Leftarrow \begin{cases} \text{Equivalent viscous damping for structural damping:} \\ c_{eq} = \dfrac{\alpha}{\pi\omega} \Leftarrow \Delta E_{cycle} = \alpha x_0^2 = \pi c_{eq}\omega x_0^2 \end{cases}$$

$$m\ddot{x} + \frac{\alpha}{\pi\omega}\dot{x} + kx = f_0 e^{j\omega t}$$

$$\downarrow\Leftarrow x(t) = x_0 e^{j(\omega t-\phi)};\ \dot{x}(t) = x_0 j\omega e^{j(\omega t-\phi)}$$

$$m\ddot{x} + j\omega\frac{\alpha}{\pi\omega}x_0 e^{j(\omega t-\phi)} + kx = f_0 e^{-j\omega t}$$

$$m\ddot{x} + \left(1 + j\frac{\alpha}{\pi k}\right)kx(t) = f_0 e^{j\omega t}$$

$$\downarrow\Leftarrow \eta\,(\gamma \text{ in text}) \equiv \frac{\alpha}{\pi k}:\ \text{loss factor (or structural damping factor)}$$

$$m\ddot{x} + \underbrace{k(1+j\eta)}_{\text{complex stiffness}}\ x(t) = f_0 e^{j\omega t}\ (\Rightarrow \text{Useful for transient(non-harmonic) excitations?})$$

Note:

$\Delta E_{cycle} = \pi k \eta x_0^2$ in structurally damped system !

Diverging response and Causality not guaranteed, hence, not good for time domain analysis! See Supplimentary Note.

3.4 Response to Periodic Excitations 2nd Order System

$$m\ddot{x} + c\dot{x} + kx = f(t), \text{ where } f(t) = f(t+T)$$

$$\downarrow\Leftarrow \begin{cases} f(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_o t}, \ \omega_o = \dfrac{2\pi}{T} \text{ for } f(t) = f(t+T) \\ \text{where } a_k = \dfrac{1}{T}\int_0^T f(t) e^{-jk\omega_o t} dt \\ x(t) = \sum_{k=-\infty}^{\infty} A_k e^{j(k\omega_o t - \phi_k)}, \end{cases}$$

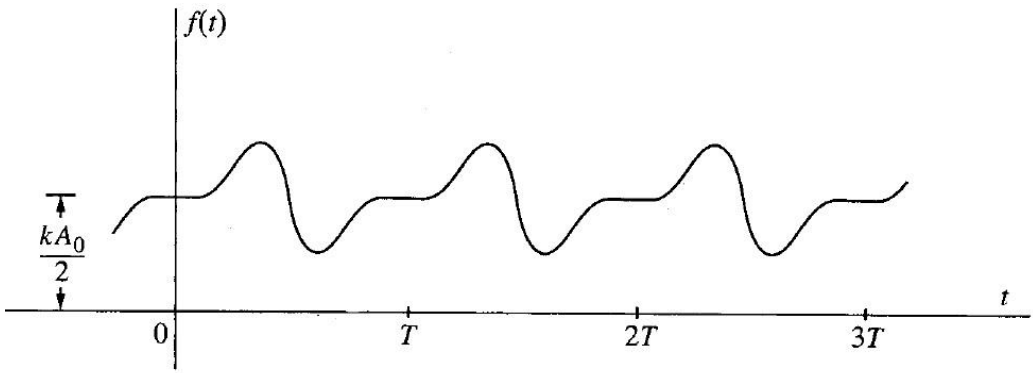

Figure 3.15 Periodic function

$$G(j\omega)\Big|_{\omega=k\omega_o} = \frac{A_k e^{-j\phi_k}}{a_k} = \frac{1}{-m(k\omega_o)^2 + jc(k\omega_o) + k} \text{ or}$$

$$x(t) = \sum_{k=-\infty}^{\infty} a_k G(j\omega)\Big|_{\omega=k\omega_o} e^{jk\omega_o t} = \sum_{k=-\infty}^{\infty} \frac{a_k}{-m(k\omega_o)^2 + jc(k\omega_o) + k} e^{jk\omega_o t}$$

Note:

- For a harmonic input with frequency ω, we have a harmonic output with frequency ω!
- How about for a square input? Will we have a square output of the same period, in cases of undamped system v.s. damped system?

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Highlight 7

- We learned about SDOF vibration system with initial conditions, harmonic excitation and periodic excitations (combination of harmonics).

- For a harmonic input with frequency ω, we have a harmonic output with frequency ω as long as the system is linear time-invariant (this is power of harmonic analysis) !

- Structural damping yields energy dissipation proportional to square of displacement, commonly treated as an equivalent viscous damping. The structural damping model is valid only in harmonic excitation and corresponding response.

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Supplementary (Optional):

Time domain analysis of structural damping system

(A failure cases: complex time response, non-causal response due to impulse)

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Time Response of Structurally Damped System due to impulse

$m\ddot{x}+k(1+j\eta)x(t)=\delta(t), x(0^-)=\dot{x}(0^-)=0$: Linear ODE with complex coefficients

$\downarrow\Leftarrow$ 1) Lapalce transform

$$ms^2X(s)+k(1+j\eta)X(s)=1: X(s)=\frac{1}{ms^2+k(1+j\eta)}=\frac{1}{m}\left(\frac{a_1}{s-p_1}+\frac{a_2}{s-p_2}\right)$$

$\downarrow\Leftarrow$ Inverse Lapalce transform

$$x(t)=\frac{1}{m}\left(a_1e^{p_1t}+a_2e^{p_2t}\right), \text{ where}$$

$$a_i=\frac{1}{2p_i}, i=1,2$$

$$p_i=\begin{cases}\pm\left(\dfrac{\sqrt{1+\eta^2}-1}{2}\right)^{\frac{1}{2}}\left(1-j\dfrac{\sqrt{1+\eta^2}+1}{\eta}\right)\omega_n \\ \text{Note that } \lim\limits_{\eta\to 0}(p_i)=\mp j\omega_n.\end{cases}$$

(You check the derivations! Note that $x(t)$ is not only complex but also the term of p_i with positive real part yields unstable solution!)

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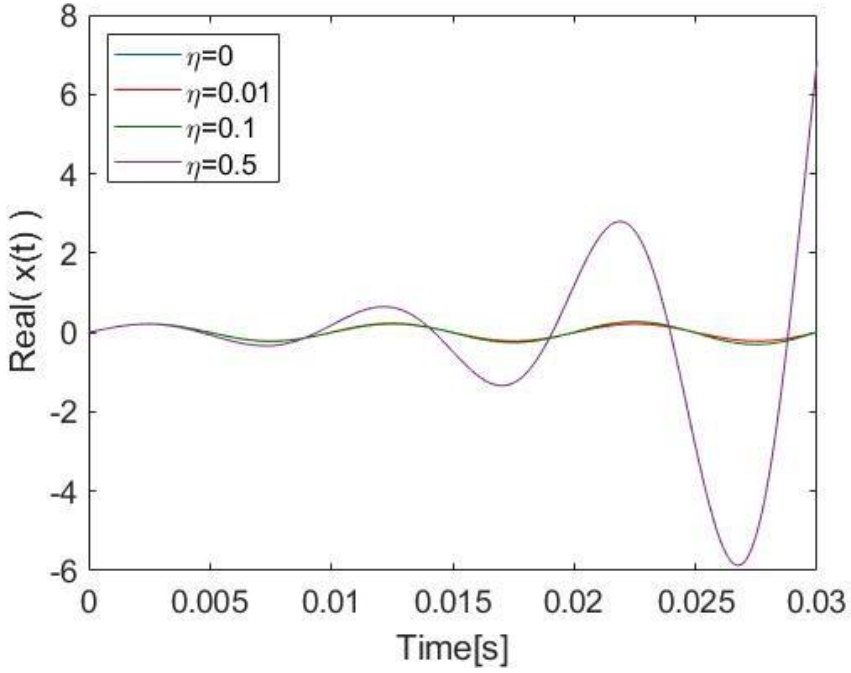
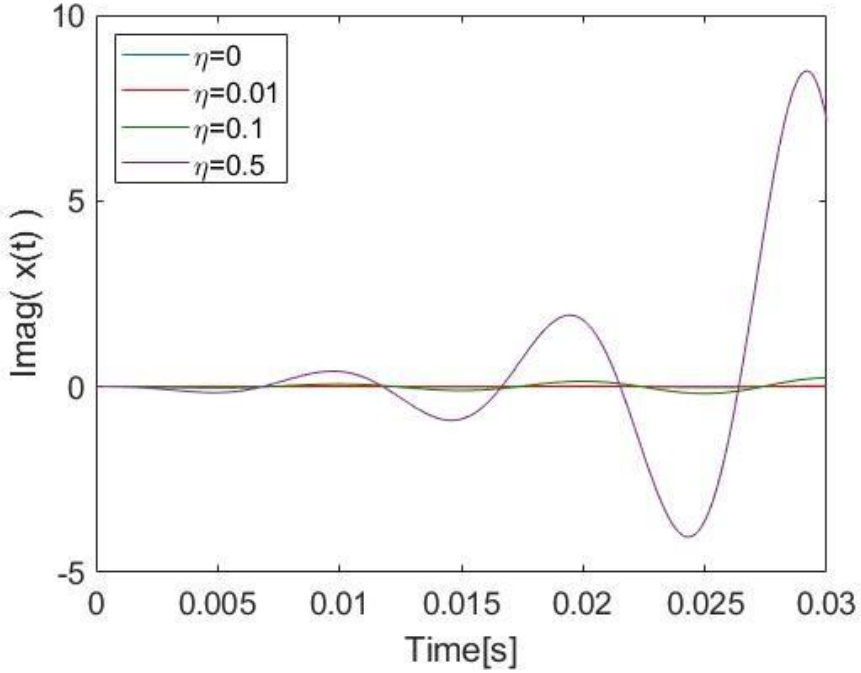
Instability of $x(t)=\frac{1}{m}\left(a_1e^{p_1t}+a_2e^{p_2t}\right)$, : Complex x(t) even with real value excitation function

where

$$a_i=\frac{1}{2p_i},p_i=\pm\left(\frac{\sqrt{1+\eta^2}-1}{2}\right)^{\frac{1}{2}}\left(1-j\frac{\sqrt{1+\eta^2}+1}{\eta}\right)\omega_n,i=1,2$$





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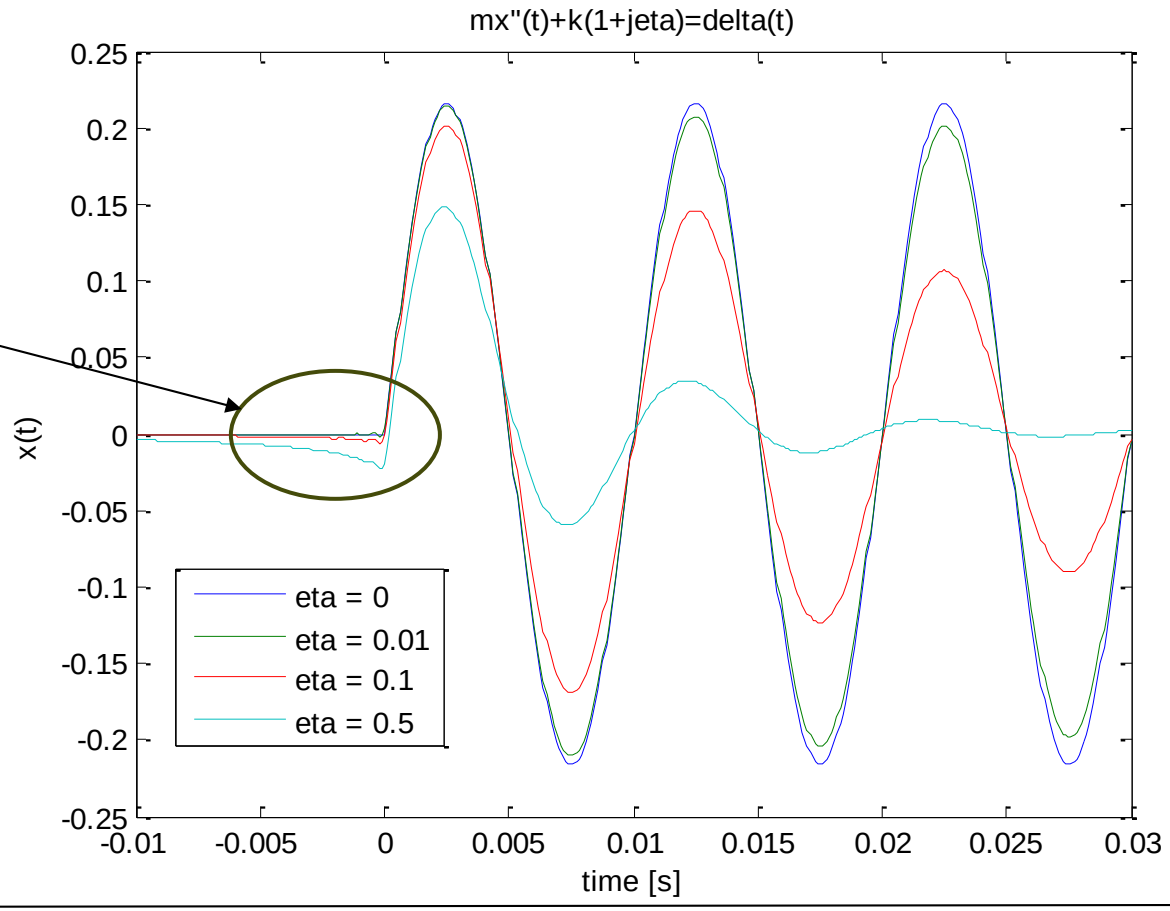
Non-causality of Structurally Damped 1-DOF System by inverse Fourier Transform

$$x(t) = F^{-1}\left[X(s)\big|_{s=j\omega}\right] = F^{-1}\left[\frac{1}{ms^2 + k(1+j\eta)}\bigg|_{s=j\omega}\right] \text{ for } m = 7.385\times10^{-3}\text{kg}, k = 2910N/m$$

$\text{Im}\left[X(-j\omega)\right] = -\text{Im}\left[X(j\omega)\right]$ is imposed to make $x(t)$ real in the simulation.

Non-causal response x(t) due by impulse at t=o

$$x(t) \neq 0 \; for \; t < 0$$



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